April 30, 2025

Letter of Support

To whom it may concern:

I am writing on behalf of NVIDIA Corporation to endorse the application "**Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools (MomEnt)"** to the **Research Council of Finland (RCF) – 2025 Special funding for international collaboration in High-Performance Computing**, led by professor Maarit Korpi-Lagg.

NVIDIA is greatly invested in developing state-of-the-art hardware and software solutions in the fields of accelerated computing and Artificial Intelligence (AI). We see physical sciences becoming a domain that can significantly benefit from AI, as demonstrated in the recent advances in AI-based weather forecasting models which can deliver accurate predictions orders of magnitude faster than their traditional counterparts. NVIDIA is also developing various tools to support the convergence of numerical simulations and AI, including NVIDIA PhysicsNeMo, an open-source deep learning framework for Physics-AI models, NVIDIA Warp, a framework for building accelerated and differentiable simulation code, and TorchFort, a C/Fortran library that allows strong coupling of AI frameworks and numerical simulation HPC codes.

The foundational research goal of MomEnt is to advance AI-accelerated solvers for plasma simulations, as classical HPC methods with increasing mesh sizes are becoming intractable due to explosion of data. The group aims to achieve this by undertaking large scale high-fidelity simulations from which the AI models can learn the interaction between complex physics at the smallest scales and physics at larger scales. When trained, they can be run in conjunction with new numerical simulations to improve their accuracy and stability at more tractable resolutions.

NVIDIA has had the opportunity to support Prof. Maarit Korpi-Lagg's research, under the umbrella of the NVIDIA AI Technology Center (NVAITC) Finland and can attest of the quality of the work carried out at the group. We look forward to collaborating with the group on the aforementioned proposal. Please note that while NVIDIA has an intent to help, any formal collaboration is subject to mutual written agreement signed by the parties and NVIDIA's plans are subject to change.

Best regards,

Marc Hamilton
Vice President, Solutions Architecture and Engineering
NVIDIA Corporation